

CLEMENT TOH

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EDUCATION

National University of Singapore (NUS)

- Bachelor of Engineering (Mechanical Engineering) with Honours
- Relevant Coursework: Thermodynamics, Circuit Analysis, Additive Manufacturing

Singapore
Expected May 2028
GPA: 4.19 / 5.00

Singapore Polytechnic (SP)

- Diploma in Mechanical Engineering (with Merit)

April 2023
GPA: 3.86 / 4.00

TECHNICAL SKILLS

- Test, Validation & Instrumentation:** EVT, Root Cause Analysis, NI DAQ (NI-9236), Oscilloscopes, Strain Gauges, Acoustic Emission Sensors, Optical QC, Class 4 Laser Safety & QC
- Systems Integration & Embedded:** M5Stack/ IIoT Edge Devices, Arduino, PLCs, EPL (Ethernet POWERLINK) Network Analysis, Electro-Mechanical Troubleshooting, Hardware – Software Integration
- Data & Software:** Python (Test Automation, Data Pipelines), C++, InfluxDB, Grafana, MATLAB, PlatformIO, GitLab
- CAD & Manufacturing:** SolidWorks, Siemens NX, Autodesk Inventor, Fusion 360, DFM, BOM, Additive Manufacturing (LPBF & FDM)

PROFESSIONAL EXPERIENCE

JD Union Pte. Ltd.

Mechatronics Engineering Intern

Singapore
June 2026 – August 2026

- Executed rigorous quality control and engineering validation testing (EVT) on Class 4 laser (LUXINAR CO2); utilized microscopy to inspect Quartz Block Head (QBH) fiber optics to prevent window failure.
- Designed a >250kg 5-tier mobile test workstation in SolidWorks — 4040 aluminium extrusion framing on heavy-duty casters, minimised vibration during sub-micron laser testing; produced DFM-compliant drawings for CNC fabrication
- Commissioned the delivered workstation by mounting its laser subsystems (IPG Ytterbium fiber laser, chiller, NANIO power supply) and routing cable and component layout to manage thermal dissipation in enclosed frame.

Advanced Remanufacturing and Technology Centre (A*STAR)

Systems Integration & Development Engineering Intern

Singapore
September 2022 – April 2023

- Designed and installed custom 13mm sensor feedthrough ports through an EOS M290 LPBF printer's chamber wall to mount Rosette strain gauges and a Kistler acoustic emission sensor, routing 12 DAQ and sensor channels in via 3D-printed (Siemens NX) PCB holders — without breaking the chamber's sealed inert-gas atmosphere.
- Specified a 200mA fast-acting fuse on the signal path to eliminate spark risk, operating below 0.1% oxygen chamber
- Built end-to-end telemetry pipeline — a 12-channel NI-9236 DAQ into an M5Stack device (C++, PlatformIO); Flux queries into InfluxDB, Grafana dashboard across 5 parameters to isolate powder-bed surface deviations to ~50µm.
- Wrote Python to tap and decode Ethernet POWERLINK traffic from the printer's Industrial PC across a 30-hour monitored print, correlating the recoater blade's position against an in-situ spectrometer to trace build anomalies.

Singapore Armed Forces (Armour Simulation Wing)

Systems Operator (National Service)

Singapore
May 2023 – May 2025

- Operated and maintained military simulation hardware across a synchronized network of training cabins, collaborating with a defense engineering firm to debug hardware-software systems to operational stability.

TECHNICAL PROJECT

3D Printer Thermal & Hardware Optimization (Ender 3 V3 SE)

June 2024

- Designed a custom print-head shroud (Fusion 360) and integrated high-performance blower fans to optimize laminar airflow, resolving thermal inconsistencies and mitigating part warpage.

LEADERSHIP & COMMUNITY INVOLVEMENT

Boon Lay Youth Network

Grassroots Leader

Singapore
January 2022 – December 2023

- Organized and directed online academic support programs for underprivileged students, coordinating 20 volunteer educators and supervising over 20 concurrent breakout tutoring sessions.